Charlie Gallagher

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Professional Summary

Innovation Engineer at Hewlett Packard Enterprise building AI agents and LLM tools, with a signed full-time offer from June 2027. Led the architecture of an internal AI tool that went from an empty repo to production approval in ten weeks and earned a Bronze Star, and co-built the innovation team's fastest proof of concept to reach the product roadmap. Co-authored two technical papers on agentic AI for HPE Innovation Days and HPE TechCon. Final-year Software Design (AI & Cloud Computing) student at TUS on track for a First Class Honours, AWS Certified AI Practitioner, and Chair of the RUN-EU Student Council.

Technical Skills

- **Languages:** Python, TypeScript, Go, Java, JavaScript, SQL, Bash
- **AI and agents:** LLM integration, prompt engineering, RAG, A2A, MCP, Google ADK, LangChain, Ollama, Pydantic, PyTorch, scikit-learn, Pandas
- **Cloud and DevOps:** Kubernetes, Docker, AWS, Azure, PostgreSQL (pgvector), GitHub Actions, Jenkins, Linux
- **Web:** React, Grommet, FastAPI, Figma
- **Ways of working:** Test-driven development, CI/CD, code review, Agile/Scrum (sprints, epics, user stories), Jira, Confluence, SonarQube, technical documentation
- **Professional:** Executive and stakeholder demos, requirements gathering, cross-site collaboration (Ireland, India, US), handover and knowledge transfer, public speaking

Work Experience

Innovation Engineer (part-time) | Innovation Engineering AI Intern *Jan 2026 – Present*

Hewlett Packard Enterprise, Hybrid Cloud & Office of the CTO, Galway

Intern Jan – Aug 2026, part-time since Sep 2026. Joining full-time as Innovation Engineer (AI & Cloud) in June 2027.

- Led the architecture and backend of an AI report generation tool, taking it from an empty repo to production approval as an internal sales tool in 10 weeks, with a working proof of concept in 3. It has since been productised by a platform engineering team.
- Stopped malformed and made-up LLM output reaching customer reports by validating every response against Pydantic schemas, and extended exports from Word to PowerPoint, PDF and Excel based on stakeholder feedback.
- Delivered a VM environment analysis feature in 48 hours after senior leadership flagged it as a priority, backed by a 100+ test suite and a Kubernetes deployment for remote testing.
- Presented the tool to visiting US senior leadership in a round-table demo that led to its production approval, and later led knowledge transfer to the engineering team in India.
- Co-built a knowledge agent for a multi-agent platform (Google ADK, A2A, MCP, RAG with FastAPI and pgvector), the innovation team's fastest proof of concept to reach the product roadmap. Led to a patent mining session and two co-authored papers for HPE Innovation Days and TechCon.
- Built two action agents (storage and pre-flight checks) for an agentic cloud migration platform, and was selected for the small team that built its demo frontend (TypeScript, React) for a major industry conference.
- Developing a storage health check tool (FastAPI, React) that validates array exports and rejects any AI-written wording that changes a number.

Crew Member *Jul 2022 – Nov 2025*

McDonald's, Athlone and Supermac's & Papa John's, Birr

- Trained new staff, managed stock and compliance logs, and coordinated large catering orders in high-volume restaurants while studying.

Projects

Small Agent Safety Evaluation | Final-year project, in progress
Python, Inspect AI, AgentHarm, Ollama, pytest

- Developing an open-source tool that pairs each harmful agent task with a harmless task using the same tools, so safety scores separate a model refusing from a model simply being unable to do the task.
- Running a case study on Qwen3.5 at 4B, 9B and 27B with reasoning on and off. Built test-first with CI, a CLI, a local dashboard and PDF reports.

bee-safe-cv | RUN-EU Short Advanced Programme, Belgium
YOLOv8, ONNX, TensorFlow Lite, Flutter, Roboflow

- Built a Flutter app with an international team that detects Varroa mites on honeybees on the phone itself. Labelled data in Roboflow, trained YOLOv8, and converted to TFLite for mobile. Presented to staff and industry partners at Howest University of Applied Sciences.

mushroom-classifier | Coursework
Python, scikit-learn, Pandas, PyQt5, Matplotlib

- Built a desktop app that predicts whether a mushroom is edible from its physical features. Used a decision tree, tested on a 25% held-out split, so every prediction can be traced, and showed accuracy and the top features on a dashboard.

distributed-inventory-manager | Coursework, team project
Python, SQLite, Pytest, Jenkins, SonarQube, Jira, Confluence

- Built a role-based command-line inventory and sales tool for a liquor store as a team, run as a full Agile project: epics and user stories in Jira, design docs in Confluence, and a Jenkins pipeline running Pytest coverage and SonarQube analysis on every change.

Education

BSc (Hons) Software Design with Artificial Intelligence for Cloud Computing *Sep 2023 – May 2027*
Technological University of the Shannon (TUS), Athlone

- On track for a First Class Honours (1.1).
- Relevant modules: Data Mining and Machine Learning, Distributed Systems, Service-Oriented Architecture, Security, Software Development for Cloud, Software Engineering, Databases, Networks.

Certifications and Awards

- **AWS Certified AI Practitioner:** Amazon Web Services, Jul 2026
- **Bronze Star Award:** Hewlett Packard Enterprise, Q2 FY2026, for the AI report tool, which was also featured in an internal HPE newsletter in March 2026
- **Gaisce and John Paul II Awards:** for leadership and community service

Leadership and Volunteering

Chair, RUN-EU Student Council *2024 – Present*

- Elected Chair in 2026 after two years on the council. Represent 15,000+ students from nine European universities to the alliance's Board of Rectors, chair council meetings, and shape its international student programmes.

Class Representative, TUS Student Council and Green Campus Committee *2023 – Present*

- Represent classmates to staff and the student council, and sit on the committee behind the university's Green Campus work.

Outreach and volunteering

- Gave a guest talk to second-year software students at ATU (2026). Debated EU policy as a European Youth Parliament delegate. Mentored young coders in Python and Scratch at CoderDojo.

Interests

Game modding, tabletop RPGs and game design, photography and video editing.